

Independence

1. In each of the following situations, calculate $P(A)$, $P(B)$, and $P(A \cap B)$. Are the events independent?

a. Flip a coin twice. A = first flip is H; B = second flip is T

$P(A) =$ $P(B) =$ $P(A \cap B) =$ Independent?

b. Flip a coin twice. A = first flip is H; B = first flip is T

$P(A) =$ $P(B) =$ $P(A \cap B) =$ Independent?

c. Roll a red die and a green die. A = first roll is 1; B = second roll is 2

$P(A) =$ $P(B) =$ $P(A \cap B) =$ Independent?

d. Roll a single die. A = roll an even number; B = roll greater than 3

$P(A) =$ $P(B) =$ $P(A \cap B) =$ Independent?

e. Roll a single die. A = roll an even number; B = roll a multiple of 3

$P(A) =$ $P(B) =$ $P(A \cap B) =$ Independent?

f. Roll a red die and a green die. A = sum of the rolls is even; B = sum of the rolls is a multiple of 3

$P(A) =$

$P(B) =$

$P(A \cap B) =$

Independent?

2. Consider three events A, B, C.

a. Use the algebra of events to write the event “exactly one of A, B, C happens” using $\cup$, $\cap$, and complements.

b. Assume $P(A) = 1/4$, $P(B) = 1/3$, $P(C) = 1/2$ and that the three events are independent. That assumption tells us several things:

- $P(A \cap B \cap C) = P(A) P(B) P(C)$
- You will not need it, but it's also true that $P(A \cap B) = P(A) P(B)$, $P(B \cap C) = P(B) P(C)$, and $P(A \cap C) = P(A) P(C)$. All four product formulas are part of independence for three events.
- We can take complements of the individual events and still get independent events. For example, A, B^c , and C^c are independent and $P(A \cap B^c \cap C^c) = P(A) P(B^c) P(C^c)$.

What is the probability that exactly one of the events happens?